Ad Compliance & Brand Safety — Submission

Project Overview

An automated video ad compliance review system. Analyzes video content through a single API call via TwelveLabs Pegasus 1.2, generating structured compliance reports. An alternative Amazon Bedrock backend is also supported (see Appendix B).

Live Demo: AWS Amplify Demo

Repository: github.com/rainny72/tlabs-ad-compliance

Demo Video: ad-compliance-demo-2x.mp4

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1. How Decisions Are Made

1.1 Single API Call-Based Analysis

Video analysis is performed through a **single API call** to the TwelveLabs Pegasus 1.2 model. All of the following information is requested in one call:

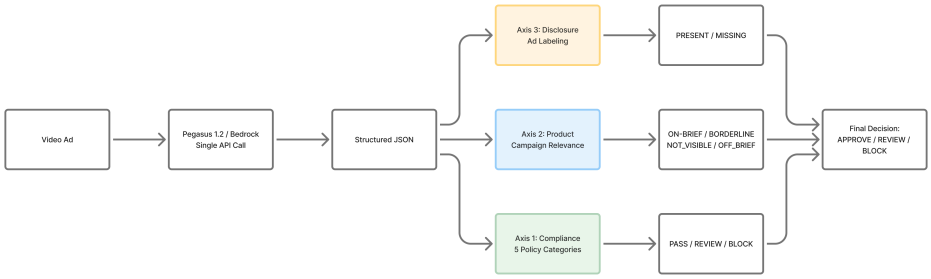
Requested Information	Description	Output Fields
Campaign Relevance	Evaluates whether the product is properly advertised	`relevance_score`, `is_on_brief`, `product_visible`

Requested Information	Description	Output Fields
Video Description	Fact-based summary of video content in 2-5 sentences	`description`
Policy Violations	Violation checks across 6 policy categories	`policy_violations` (per-category `severity`, `violations[]`)

The key benefits of this design are that the video is processed only once, keeping API costs linear, and analyzing within a single context maintains consistency across categories.

1.2 Three-Axis Evaluation System

The AI model's response is evaluated across three independent axes:



3-Axis Evaluation System

Axis 1: Compliance — Checks violations across 5 policy categories (`hate_harassment`, `profanity_explicit`, `drugs_illegal`, `unsafe_misleading_usage`, `medical_cosmetic_claims`). Each violation is assigned a severity (CRITICAL/HIGH/MEDIUM/LOW/NONE), and the highest severity among all detected violations determines the compliance result.

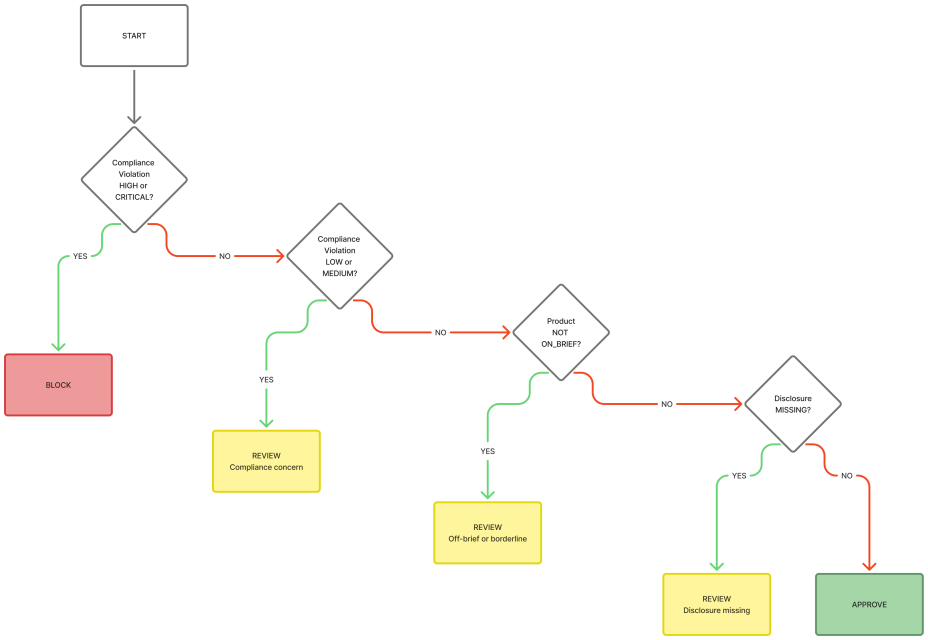
Severity	Compliance Result
CRITICAL / HIGH	BLOCK
MEDIUM / LOW	REVIEW
NONE	PASS

Axis 2: Product Relevance — Evaluates a combination of the AI model's returned `relevance_score` (0.0–1.0), `is_on_brief`, and `product_visible` flags. Scores below the 0.5 threshold are classified as BORDERLINE/OFF_BRIEF, triggering a REVIEW.

Axis 3: Disclosure — Checks for the presence of ad disclosure labels (`#ad` , `#sponsored` , etc.). Detection uses three methods: the model's explicit violation flag, severity-only detection (severity assigned without detailed violations), and keyword analysis of the video description text.

1.3 Final Decision Rules

Decision priority is BLOCK > REVIEW > APPROVE:

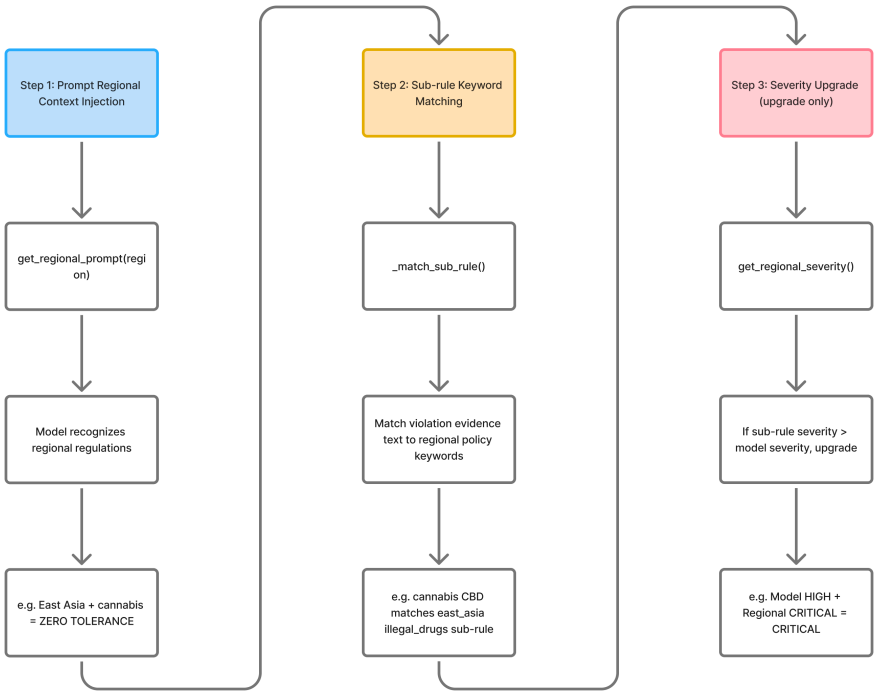


Final Decision Logic

Multiple REVIEW reasons are combined (e.g., "REVIEW: Product: OFF-BRIEF | Disclosure: MISSING").

1.4 Three-Step Regional Severity Calibration

A three-step calibration mechanism is applied to reflect regional regulatory differences. The key design insight is the separation of responsibilities: the AI model is responsible for detecting what is in the video (fact extraction), while pre-researched regional policy code is responsible for determining how severe it is in each region (regulatory judgment).



Regional Severity Calibration

Step 1: Prompt Regional Context Injection (Guide) — `get_regional_prompt(region)` injects region-specific regulatory context into the prompt, providing the model with severity classification guidance. For example, the East Asia context includes "ZERO TOLERANCE — ANY drug reference including cannabis, CBD → severity CRITICAL." This guides the model's severity assignment, but the system does not rely on the model following this guidance perfectly — Steps 2 and 3 act as a safety net.

Step 2: Sub-rule Keyword Matching (Code-Level Detection) — `_match_sub_rule()` matches the model's returned evidence text against pre-researched regional policy keywords defined in `shared/regional_policies/`. Each region's policy file contains sub-rules with specific keywords compiled from actual regulatory research (e.g., Korea MFDS, Japan Yakujiho, China NMPA). This is deterministic code-level matching — it does not depend on the model's regulatory knowledge.

Step 3: Severity Upgrade (Safety Net, upgrade only, never downgrade) — `get_regional_severity()` compares the matched sub-rule's severity with the model's returned severity. If the pre-researched policy severity is higher, it upgrades. This is strictly upgrade-only — the model's severity is never lowered, because in compliance it is safer to over-flag than to miss a violation.

Concrete Example — Drug Reference Video in East Asia Region

The video `06-mw-230-jp-stick-drugs.mp4` (a Japanese cosmetics ad containing cannabis/CBD ingredient references) demonstrates the three-step calibration in action:

Step	What Happens	Input	Output
1. Prompt Guide	East Asia context appended: "ZERO TOLERANCE… cannabis, CBD → CRITICAL"	`get_regional_prompt("east_asia")`	Model receives severity guidance
2. Model Detection	Pegasus 1.2 analyzes video, detects drug-related content in speech/text	Video multimodal analysis	`drugs_illegal: severity "high"`, evidence: "cannabis CBD hemp extract ingredient"
3a. Sub-rule Match	Evidence text matched against East Asia policy keywords	`_match_sub_rule(DRUGS_ILLEGAL, "cannabis CBD hemp extract ingredient", east_asia_policy)`	Matched sub-rule: `illegal_drugs` (keywords: "cannabis", "CBD", "大麻", "마약")
3b. Severity Upgrade	Sub-rule severity CRITICAL > model severity HIGH → upgrade	`get_regional_severity(DRUGS_ILLEGAL, "illegal_drugs", east_asia_policy)` → CRITICAL	Final severity: **CRITICAL** (upgraded from HIGH)

In this example, the model correctly detected the drug-related content (Step 2 — this is the model's strength as a video understanding model), but assigned `severity: high` instead of the East Asia-required CRITICAL. The code-level safety net (Steps 3a-3b) caught this underestimation and upgraded the severity based on pre-researched East Asia regulations where all drug references carry zero tolerance.

Same Video, Different Regions — Severity Comparison:

Region	Model Returns	Sub-rule Match	Policy Severity	Final Severity	Decision
North America	`high`	`cannabis_cbd`	HIGH	HIGH (no change)	BLOCK
Western Europe	`high`	`cannabis_cbd`	HIGH	HIGH (no change)	BLOCK
East Asia	`high`	`illegal_drugs`	CRITICAL	**CRITICAL** (upgraded)	BLOCK
Global	`high`	strictest across all	CRITICAL	**CRITICAL** (upgraded)	BLOCK

This demonstrates the core value of the calibration mechanism: the same model output produces different final severities depending on the region, reflecting actual regulatory differences. The model only needs to detect the content accurately — the regional severity judgment is handled by pre-researched policy code.

Why This Design — Model Detection + Code-Level Regulation:

- Pegasus 1.2 is a video understanding model optimized for multimodal content analysis (visual, speech, text). It excels at detecting what is in the video — drug references, profanity, unsafe usage, etc.
- However, mapping detected content to region-specific regulatory severity requires domain expertise that a video understanding model may not reliably possess (e.g., “cannabis is HIGH in North America but CRITICAL in East Asia”).
- The three-step design leverages each component’s strength: the model detects content (its core capability), the prompt provides severity guidance (best-effort), and the code enforces pre-researched regional regulations (guaranteed accuracy).
- The upgrade-only constraint ensures that even if the model overestimates severity (e.g., returns CRITICAL when the region only requires HIGH), the system preserves the model’s judgment — erring on the side of caution.

The Global region applies `get_strictest_severity()` across all regional policies, ensuring that Global-safe content is safe in every market.

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2. Why Outputs Are Trustworthy

Output trustworthiness is ensured through three pillars: consistent response generation, enforced structured output, and multi-layer verification mechanisms.

2.1 API Parameter Settings for Deterministic Output

The TwelveLabs Analyze API is configured with parameters optimized for compliance analysis:

Parameter	Value	Purpose
`temperature`	`0.1`	Most deterministic setting per TwelveLabs Temperature Tuning Guide
`max_tokens`	`4096`	Pegasus maximum to prevent truncation
`response_format`	`{type: "json_schema", json_schema: ...}`	Enforces structured output
`stream`	`false`	Complete response in single payload

The `temperature: 0.1` setting is the most deterministic option recommended by the TwelveLabs Temperature Tuning Guide for “law enforcement, reports” use cases. Compliance analysis requires the same video to produce reproducible results, making minimal non-determinism essential. The single API call design means non-determinism occurs only once, and cross-referencing across categories within a single context ensures consistent analysis results.

2.2 JSON Schema-Based Structured Output

The core principle from TwelveLabs’ Structured Responses guide is applied:

> “The schema takes precedence over the prompt.” — When the schema and prompt conflict, the schema wins.

Following this principle, `COMBINED_JSON_SCHEMA` enforces the output structure:

- All 6 policy categories are marked as `required` to prevent omissions
- `severity` is constrained to `enum [none|low|medium|high|critical]` to exclude arbitrary values
- `modality` is restricted to `enum [visual|speech|text_on_screen]` to clarify detection channels
- `relevance_score` is defined as `number (0.0-1.0)` to ensure quantitative evaluation

The schema and prompt output instructions are precisely aligned so the model follows the schema while reflecting the prompt's intent.

2.3 Prompt Architecture — COMBINED_PROMPT Design

The system uses a single unified prompt (`COMBINED_PROMPT`) that instructs the model to perform all analysis tasks in one pass. The prompt is composed of four structural layers, each serving a distinct purpose.

Layer 1: Role Definition and Severity Judgment Guide

The prompt opens by assigning the model a specific expert role with named regulatory bodies:

> "You are an expert ad compliance reviewer specializing in global beauty and cosmetics advertising regulations, including FTC (US), ASA (UK), EU Cosmetics Regulation, and East Asian standards (Korea MFDS, Japan Yakujiho, China NMPA)."

Instead of a generic "content moderator," naming specific regulatory bodies provides the model with a structured severity judgment framework. Pegasus 1.2 is a video understanding model optimized for multimodal analysis (visual, speech, text), not a general-purpose LLM with deep regulatory knowledge. Therefore, the prompt's regulatory references serve as severity classification guides rather than activating pre-existing domain expertise. The system does not rely solely on the model's regulatory understanding — the three-step calibration mechanism (Section 1.4) ensures accurate regional compliance through code-level sub-rule matching (`_match_sub_rule()`) and severity upgrade (`get_regional_severity()`), which act as a safety net for any gaps in the model's domain knowledge.

Layer 2: Multi-Modality Analysis Instruction

The prompt explicitly requires independent analysis of all three modalities before making any judgment:

> "Analyze ALL three modalities independently before making any judgment: 1. VISUAL: actions, products, text overlays, subtitles 2. SPEECH/AUDIO: transcribe ALL spoken words verbatim in any language 3. TEXT ON SCREEN: all overlays, captions, hashtags, watermarks

> Do NOT assume a video is clean based on visuals alone. Violations often appear only in audio or text."

This instruction is critical because compliance violations frequently appear in only one modality — profanity in speech, drug references in on-screen text, or unsafe usage in visual content. Without this directive, the model may default to visual-only analysis.

Layer 3: Three Output Sections with Explicit Instructions

The prompt defines three output sections that map directly to the JSON Schema:

1. **Campaign Relevance** — Instructs the model to score relevance (0.0–1.0), determine if the product is on-brief, and check product visibility. These fields feed directly into Axis 2 (Product Relevance) of the evaluation system.

2. **Video Description** — Requires a factual 2-5 sentence summary with verbatim transcription of all spoken dialogue (original language + English translation), all on-screen text, and any offensive language exactly as spoken. The description serves dual purposes: it provides human-readable context for reviewers, and it feeds into the keyword-based disclosure detection path (Axis 3).

3. **Policy Violations** — Defines 6 categories with per-category severity mapping. Each category includes specific examples and expected severity levels:

Category	Key Severity Mappings in Prompt
`hate_harassment`	Racial slurs → critical, skin tone superiority → critical, body shaming → high
`profanity_explicit`	Strong profanity (F-word, 씨발, くそ, 他妈的) → critical, mild → medium
`drugs_illegal`	Illegal drug use → critical, cannabis/CBD → high (critical in East Asia), substance-derived cosmetic ingredients → high
`unsafe_misleading_usage`	Unsafe application → high, misleading before/after → high
`medical_cosmetic_claims`	Drug claims ("cures", "treats") → critical, "FDA approved" for cosmetics → critical, absolute claims → high
`disclosure`	No disclosure at all → medium, buried disclosure → low

The per-category severity mapping is essential for consistency — it ensures the model returns the same severity for identical violations across different videos, rather than making ad-hoc judgments.

Layer 4: Regional Context Injection (Dynamic)

The base `COMBINED_PROMPT` is extended at runtime by `get_regional_prompt(region)`, which appends region-specific regulatory context from `REGIONAL_PROMPT_CONTEXT`. Four regional contexts are defined:

Region	Key Directives	Regulatory Bodies
`north_america`	CBD → HIGH, "FDA approved" for cosmetics → CRITICAL, FTC disclosure rules	FTC, FDA, MoCRA
`western_europe`	ASA strictest disclosure rules ("#sponsored" is inadequate → HIGH), retouched before/after banned	ASA, ARPP, EU Cosmetics Reg
`east_asia`	ZERO TOLERANCE for any drug reference → CRITICAL, mild profanity → HIGH, absolute claims → CRITICAL	Korea MFDS, Japan Yakujiho, China NMPA
`global`	No additional context (base prompt applies strictest defaults)	—

The regional context is appended as a `## REGIONAL CONTEXT` section, not replacing the base prompt. This means the model always has the full base knowledge plus region-specific overrides. For example, the East Asia context explicitly states "ANY drug reference including cannabis, CBD, marijuana → severity CRITICAL (not high)" — overriding the base prompt's default of HIGH for cannabis.

API Prompt Token Limit and Design Constraint

The TwelveLabs Analyze API (`POST /v1.3/analyze`) and Amazon Bedrock `invoke_model` both impose a maximum prompt length of 2,000 tokens (`inputPrompt` / `prompt` parameter). This constraint directly shaped the prompt design — all 6 policy categories, severity mappings, multi-modality instructions, and output format specifications must fit within this budget.

Prompt Variant	Estimated Tokens	Budget Usage
Global (base only)	~914	46%
North America	~1,088	54%
Western Europe	~1,085	54%
East Asia (longest)	~1,137	57%

The prompt was designed following the [TwelveLabs Prompt Engineering Guide]() best practices: role definition with specific context, explicit output format specification, concise and specific instructions, and temperature tuning for deterministic compliance output. The per-category severity mappings are kept concise (arrow notation: `"FDA approved" for cosmetics -> critical`) rather than verbose explanations, specifically to stay within the 2,000-token budget while maximizing the information density delivered to the model.

Why This Prompt Structure Works

The four-layer design ensures: (1) the model receives structured severity judgment guidance through named regulatory contexts, (2) it analyzes all modalities rather than defaulting to visual-only, (3) it produces consistent severity assignments through explicit per-category mapping, and (4) it respects regional regulatory differences through dynamic context injection. The prompt and JSON Schema are precisely aligned — the schema enforces structure while the prompt provides the judgment criteria. Importantly, the system does not depend on the model's inherent regulatory knowledge — the prompt guides severity classification, while code-level calibration (Section 1.4) guarantees accurate regional enforcement.

2.4 Multi-Layer Verification Mechanisms

Model output trustworthiness is further verified at the code level:

Truncation Detection — The API response's `finishReason` field is checked; if `"length"`, a warning is raised that the response was truncated. Truncated responses may contain incomplete JSON and are immediately flagged.

Regional Policy Safety Net — Even if the model underestimates regional regulations (e.g., returning `severity: high` for cannabis in East Asia instead of CRITICAL), code-level `_match_sub_rule()` + `get_regional_severity()` matches evidence text against pre-researched regional policy keywords and upgrades severity. This calibration is upgrade-only (never downgrade), preventing false negatives. The regional policies in `shared/regional_policies/` are compiled from actual regulatory research (Korea MFDS, Japan

Yakujiho, China NMPA, FTC, ASA, EU Cosmetics Regulation), ensuring that the system's regulatory accuracy does not depend on the model's domain knowledge.

Multi-Path Disclosure Detection — Disclosure detection does not rely on a single method but uses three paths:

1. The model's explicit violation flag
2. Severity-only detection (severity assigned without detailed violations)
3. Keyword analysis of the video description text ("no disclosure", "missing #ad", etc.)

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3. How This Would Scale in a Real Ads System

3.1 Scale Requirements of Real Ad Systems

Large ad platforms must process hundreds of thousands to millions of ad content items per day. This creates the following scaling requirements:

Requirement	Current System	Production Scale
Daily throughput	Tens of items (demo)	Hundreds of thousands to millions
Response time	10-60s per item	SLA-based (tiered by priority)
Regional policies	4 regions (code)	Dozens of regions (Policy DB)
Review workflow	Dashboard display	Tiered auto-routing
Monitoring	CloudWatch Logs	Real-time metrics + drift detection

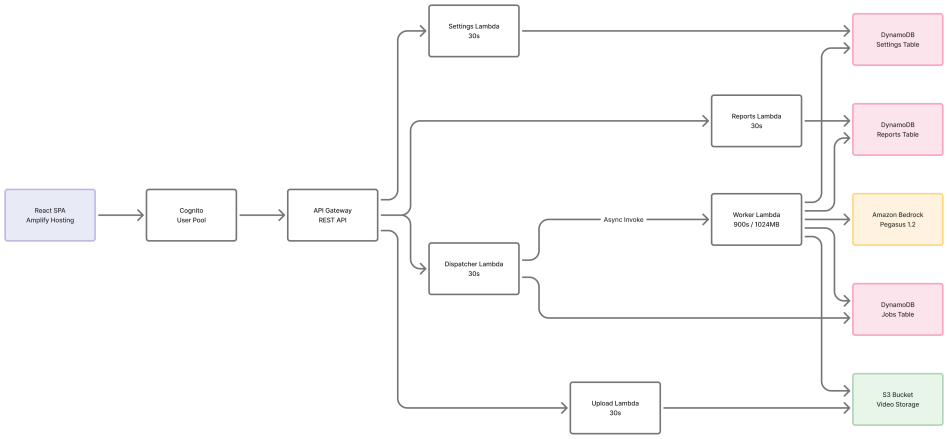
3.2 Scaling Architecture Design

The current system's core design principles naturally support large-scale expansion:

Single API Call = Linear Cost Scaling — All analysis is completed in one call per video, so when throughput increases 10x, cost increases only 10x. This is 8x more cost-efficient compared to individual calls per category (8 calls).

Stateless Workers = Horizontal Scaling — Workers hold no state, so throughput can be scaled simply by increasing the number of concurrent executions. There is zero coordination overhead between workers.

Policy as Code = No Model Retraining Required — When new regions or regulations are added, only policy files need to be added. There is no need to retrain or redeploy the AI model.



Production Architecture

3.3 Tiered Processing Strategy

In real ad systems, not all content is treated equally. Processing is differentiated based on the decision result:

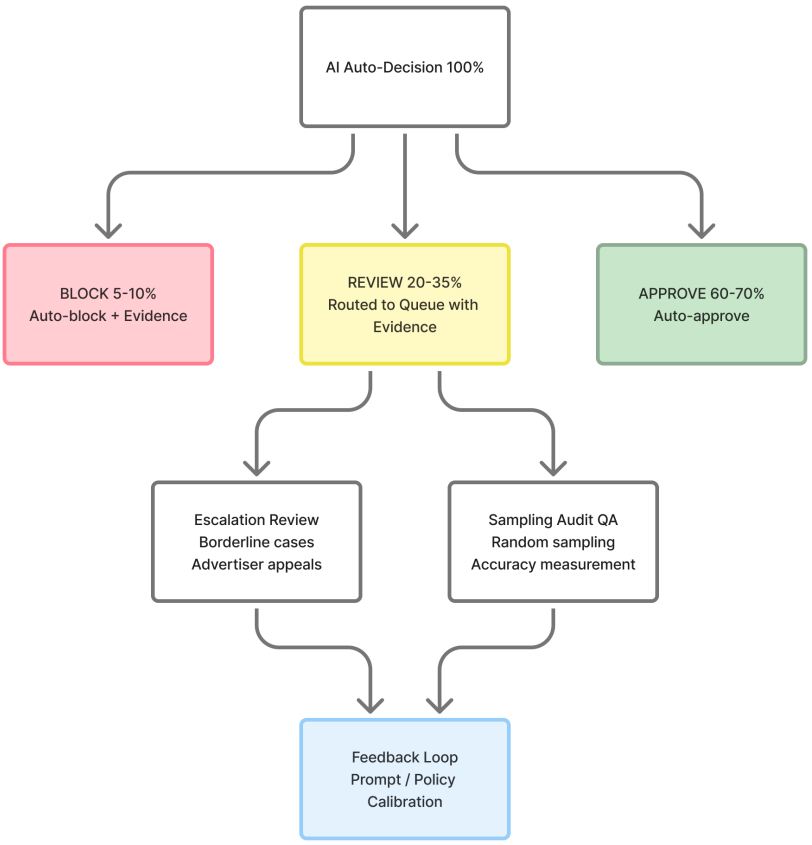
Tier	Decision	Processing	Expected Ratio
Auto-approve	APPROVE (high confidence)	Automatic approval, no human review needed	60-70%
Auto-block	BLOCK (CRITICAL severity)	Automatic block, advertiser notification	5-10%
Human review	REVIEW	Routed to reviewer queue	20-35%

This tiered classification reduces the human reviewer burden by 60-70% while immediately blocking dangerous content.

3.4 Quality Control and Human-in-the-Loop

The system's core goal is to automate the Ads Compliance team's review process. To handle hundreds of thousands of ads per day, AI must perform first-pass decisions

automatically and provide clear decision rationale. Human-in-the-Loop serves not as a full review but as a sampling-based audit to continuously verify and improve model quality.



Human-in-the-Loop Quality Control

Automated Decision + Decision Rationale — For BLOCK/REVIEW decisions, the violation category, severity, and evidence (detection modality and specific content) are provided in a structured format. This fulfills the “Clear explanations for any blocked or reviewed ads” requirement, allowing advertisers to immediately see the reason for blocking/review.

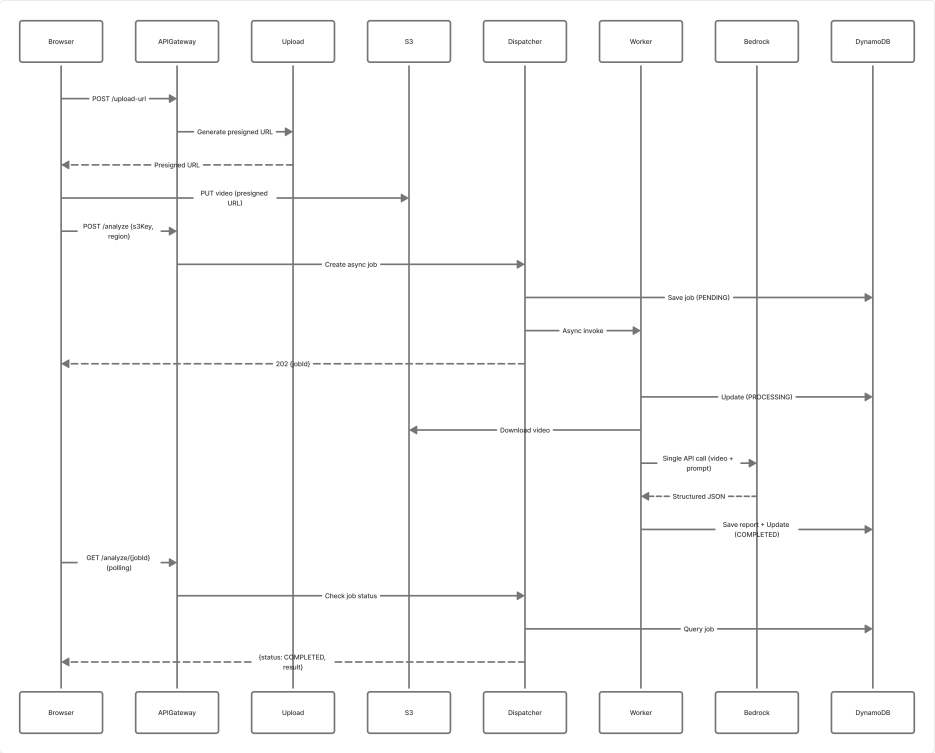
Sampling-Based Quality Audit — A percentage of all decisions is randomly sampled for human auditor verification. False positive and false negative rates are measured, and when accuracy for a specific category falls below the threshold, prompts or regional policies are adjusted.

Drift Detection — Changes in decision distribution (APPROVE/REVIEW/BLOCK ratios) over time are tracked. Sudden distribution shifts indicate model behavior changes or new content types, signaling when policy adjustments are needed.

Regional Policy Updates — When regulations change, only policy files need to be updated for immediate effect. Extending code-based policy management (`shared/regional_policies/`) to a Policy DB + Admin UI would allow compliance officers to manage policies directly.

3.5 Serverless Architecture-Based Scaling Direction

Based on the current demo system's serverless structure, the architecture can evolve as follows for large-scale deployment:



Video Analysis Sequence

- **SQS Queue Introduction**: Instead of the Dispatcher directly invoking the Worker, messages are published to SQS for automatic backpressure control and retry handling
- **Reserved Concurrency**: Worker Lambda concurrent execution limits prevent exceeding Bedrock API call quotas
- **Dead Letter Queue**: Failed analyses are stored in a DLQ for reprocessing or manual review

- **Step Functions**: Complex workflows (analysis -> review routing -> notification) are orchestrated

The key advantage of this serverless architecture is automatic scaling up/down based on traffic with no infrastructure management burden, paying only for what is used.

Appendix A: Production Implementation Example

> A production deployment was implemented to validate the design thinking. For details on system architecture, security design, and CDK infrastructure, see the document below.

Detailed documentation: Application Architecture — Includes dual deployment structure (Streamlit/Amplify), Dispatcher/Worker async pattern, CDK infrastructure stacks, Lambda function structure, and security design (Cognito authentication, S3 public access blocking, least privilege principle).

Appendix B: Dual Backend Architecture

The system supports two interchangeable backends for video analysis. Users can switch between them via the Settings page, and the Worker Lambda routes to the selected backend at runtime. Both backends share the same prompt (`get_regional_prompt(region)`), JSON schema (`COMBINED_JSON_SCHEMA`), and decision engine (`make_split_decision`), ensuring identical evaluation logic regardless of which backend processes the video.

Aspect	TwelveLabs Direct API	Amazon Bedrock
Endpoint	`POST /v1.3/analyze`	`invoke_model` (bedrock-runtime)
Model	Pegasus 1.2 (hosted by TwelveLabs)	`twelvelabs.pegasus-1-2-v1:0` (AWS Marketplace)
Authentication	API Key (`x-api-key` header)	AWS IAM (Signature V4)
Video Input	Indexed asset (upload → index → analyze)	Base64 or S3 URI

Aspect	TwelveLabs Direct API	Amazon Bedrock
Max Video Size	Managed by TwelveLabs indexing pipeline	25 MB (base64) / 2 GB (S3 URI)
Temperature	`0.1`	`0.1`
Best For	TwelveLabs-native workflows, rapid prototyping	Enterprise AWS environments, IAM-governed access

TwelveLabs Direct API Pipeline (4 Steps):

1. **Index Creation** — `POST /v1.3/indexes` creates a named index (`ad-compliance-analysis`) with `pegasus1.2` model and `["visual", "audio"]` modalities.
2. **Asset Upload** — `POST /v1.3/assets` uploads the video via multipart/form-data. Polls until `"ready"`.
3. **Asset Indexing** — `POST /v1.3/indexes/{index_id}/indexed-assets` links the asset to the index. Polls until `"ready"`.
4. **Analysis** — `POST /v1.3/analyze` sends the prompt, JSON schema, and parameters. This single call produces the structured compliance report.

Amazon Bedrock Pipeline (Single Step): Bedrock simplifies to a single `invoke_model` call. The video is passed as base64 (≤25 MB) or S3 URI (≤2 GB). No pre-indexing required.